

Section : Physics

Q.1 To increase the current sensitivity of a moving coil galvanometer by 50%, its resistance is increased so that the new resistance becomes twice its initial resistance. By what percentage does its voltage sensitivity change?

1. It increases by 75%.
2. It decreases by 75%.
3. It increases by 25%.
4. It decreases by 25%.

Options

1. 1
2. 2
3. 3
4. 4

Q.2 A wire has a resistance of 3.0Ω at 30°C and a resistance of 4.5Ω at 100°C . The temperature coefficient of resistance of the wire is

1. $0.0032^\circ\text{C}^{-1}$
2. $0.0071^\circ\text{C}^{-1}$
3. 0.012°C^{-1}
4. 0.126°C^{-1}

Options

1. 1
2. 2
3. 3
4. 4

Q.3 The maximum torque acting on a coil having an effective area 0.04 m^2 and carrying current of $100\mu\text{A}$ is $4 \times 10^{-8}\text{ N m}$, when placed in a magnetic field. The magnitude of the magnetic field is

1. 10^{-1} Wb m^{-2}
2. 10^{-2} Wb m^{-2}
3. $2 \times 10^{-3}\text{ Wb m}^{-2}$
4. $4.2 \times 10^{-2}\text{ Wb m}^{-2}$

Options

1. 1
2. 2
3. 3
4. 4

Q.4 The power of a biconvex lens is 12 D and the radius of curvature of each surface is 10 cm. The refractive index of the material of the lens is

1. 1.6
2. 1.5
3. 1.45
4. 1.875

Options

1. 1
2. 2
3. 3
4. 4

Q.5 The electric current in a circular coil of 3 turns produces a magnetic field B_1 at its centre. The coil is unwound and is rewound into a circular coil of 6 turns. Now on passing the same current, the magnetic field produced is B_2 at its centre. The ratio $B_1 : B_2$ is

1. 1 : 1
2. 1 : 2
3. 1 : 4
4. 4 : 1

Options

1. 1
2. 2
3. 3
4. 4

Q.6 In a transformer, the input current and voltage are 3 A and 50 V respectively. The ratio of the number of turns in the primary coil to secondary coil is 5 : 3. The output current and voltage, respectively, are

1. 5 A and 20 V
2. 6 A and 30 V
3. 4 A and 50 V
4. 5 A and 30 V

Options

1. 1
2. 2
3. 3
4. 4

- Q.7** Forward biasing of a p-n junction
1. increases its resistance
 2. decreases its resistance
 3. increases the junction field
 4. increases the potential barrier

Options

1. 1
2. 2
3. 3
4. 4

- Q.8** Identify the correct statements.
- (A) The density of nuclear matter is independent of the size of the nucleus.
- (B) Nuclear forces are charge independent and are non-central forces.
- (C) The nuclides having the same mass number are called isotones.
- (D) The mass of a nucleus can be less or more than the sum of the masses of nucleons present inside it.
- Choose the correct answer from the options given below:
1. (A), (C) and (D) only
 2. (A) and (B) only
 3. (B) and (C) only
 4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

- Q.9** The current and voltage in an a. c. circuit are given as
- $$I = 50 \sin(50t) \text{ mA}$$
- $$V = 100 \sin(50t + \pi/3) \text{ volt}$$
- The power dissipated in the circuit is
1. 1.25 W
 2. 2.50 W
 3. 1250 W
 4. 10 W

Options

1. 1
2. 2
3. 3
4. 4

- Q.10** If the two particles of the same mass and same charge are accelerated with a potential difference of 120 V and 480 V respectively, the ratio of de Broglie's wavelengths associated with the two particles respectively is
1. 1:2
 2. 2:1
 3. 1:4
 4. 4:1

Options

1. 1
2. 2
3. 3
4. 4

- Q.11** An electric dipole of length 4 cm, when placed with its axis making an angle of 60° with a uniform electric field, experiences a torque of $4\sqrt{3} \text{ N m}$. If the dipole has charges of $\pm 8 \text{ nC}$, then the potential energy of the dipole in the electric field is:
1. $1.25 \times 10^{10} \text{ J}$
 2. $3.8 \times 10^2 \text{ J}$
 3. -4 J
 4. -16 J

Options 1. 1

2. 2

3. 3

4. 4

Q.12 When a current carrying coil is placed in a uniform external magnetic field with its magnetic moment antiparallel to the external magnetic field, then

(A) its potential energy is minimum

(B) its potential energy is maximum

(C) torque on it is maximum

(D) torque on it is zero

(E) coil is in unstable equilibrium

Choose the correct answer from the options given below:

1. (A), (C) and (E) only

2. (B), (D) and (E) only

3. (B), (C) and (E) only

4. (A), (D) and (E) only

Options 1. 1

2. 2

3. 3

4. 4

Q.13 A thick long straight wire of radius 'a' is carrying steady current I . The current I is uniformly distributed across the cross-section. The ratio of magnetic fields at a distance $a/2$ and $3a/2$ from the central axis of the wire is

1.2:3

2.3:2

3.3:4

4.4:3

Options 1. 1

2. 2

3. 3

4. 4

Q.14 Two parallel beams of monochromatic light pass through a long slit of width 0.2 mm one by one. The angular divergence in which most of the light is diffracted for both the beams are 2.25×10^{-3} rad and 4.50×10^{-3} rad respectively. If the wavelength of the first beam is $\lambda_1 = 3000 \text{ \AA}$, then the wavelength of the second beam is:

1.3000 $\text{\AA}$

2.4500 $\text{\AA}$

3.6000 $\text{\AA}$

4.7500 $\text{\AA}$

Options 1. 1

2. 2

3. 3

4. 4

Q.15 A magnetic field B exists in a direction perpendicular to the plane of a square loop made of metal wire. The wire has a diameter of 4 mm and a total length of 30 cm. The magnetic field B changes at a steady rate of 0.032 T s^{-1} . The induced current in the closed loop is

(Given: Resistivity of the metal wire is $\rho = 1.57 \times 10^{-8} \Omega \text{ m}$)

1.0.34 A

2.0.48 A

3.0.53 A

4.0.61 A

Options 1. 1

2. 2

3. 3

4. 4

- Q.16** The magnetic field in a plane electromagnetic wave is given by $B_y = 2 \times 10^{-7} \sin(0.5 \times 10^3 x + 1.57 \times 10^{11} t)$ T. The frequency of the wave is
1. 1.23.9 GHz
 2. 2.25.0 MHz
 3. 3.25.0 GHz
 4. 4.23.9 THz

Options

1. 1
2. 2
3. 3
4. 4

- Q.17** The current in a device varies with time 't' as $i = 6t$, where i is in mA and t is in s. The amount of charge that passes through the device during $t = 0$ s to $t = 3$ s is
1. 10 mC
 2. 18 mC
 3. 27 mC
 4. 54 mC

Options

1. 1
2. 2
3. 3
4. 4

- Q.18** A charge of $9 \mu\text{C}$ is given a displacement of $\vec{r} = (4\hat{i} + 5\hat{j})$ m in an electric field $\vec{E} = (5\hat{i} + 4\hat{j})$ N/C. The work done in the process is
1. 1.45×10^{-4} J
 2. 2.90×10^{-4} J
 3. 3.18×10^{-4} J
 4. 4.36×10^{-4} J

Options

1. 1
2. 2
3. 3
4. 4

- Q.19** A radioactive substance is emitting β^- particles at a certain rate. When it is heated to a high temperature, the rate of emission
1. increases
 2. decreases
 3. remains same
 4. initially increases and then decreases

Options

1. 1
2. 2
3. 3
4. 4

- Q.20** A radio wave of wavelength 300 m can be transmitted by a transmission center. A condenser of capacity $2.5 \mu\text{F}$ is available. The inductance of the required coil at resonance is
(Given: $\pi^2 = 10$)
1. 1.100×10^{-8} H
 2. 2.212×10^{-8} H
 3. 3.100×10^8 H
 4. 4.212×10^8 H

Options

1. 1
2. 2
3. 3
4. 4

- Q.21** To demonstrate the phenomenon of interference, we require two sources, which emit radiation of
1. nearly the same frequency
 2. the same frequency
 3. different wavelength
 4. the same frequency and having definite phase relationship

Options

1. 1

- 2. 2
- 3. 3
- 4. 4

Q.22 Which of the following statements are correct?

- (A) The emf of a cell depends on the internal resistance of the cell.
- (B) The emf of a cell depends on the external resistance of the circuit.
- (C) It is easier to start a car engine on a warm day than on a cold day.
- (D) The temperature coefficient of resistance (α) of carbon is negative.

Choose the correct answer from the options given below:

- 1. (A), (B) and (C) only
- 2. (C) and (D) only
- 3. (B), (C) and (D) only
- 4. (A), (B) and (D) only

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.23 Three capacitors each of capacitance $3\ \mu\text{F}$ are provided. These cannot be combined to provide the resultant capacitance of:

- 1. $1\ \mu\text{F}$
- 2. $2\ \mu\text{F}$
- 3. $4.5\ \mu\text{F}$
- 4. $6\ \mu\text{F}$

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.24 A $100\ \mu\text{F}$ capacitor is charged with a 50 V source supply. Then the source supply is removed, and the capacitor is connected across a pure inductor coil. As a result of which 5 A current flows through the inductor. The value of the self-inductance of the coil is:

- 1. 0.01 H
- 2. 0.02 H
- 3. 0.1 H
- 4. 0.17 H

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.25 Identify the correct statements.

- (A) Photographic plates are sensitive to infrared waves.
- (B) Infra-red waves are invisible but can cast shadows like visible light.
- (C) Photographic plates are sensitive to ultraviolet rays.
- (D) Infrared photons have more energy than photons of visible light.

Choose the correct answer from the options given below:

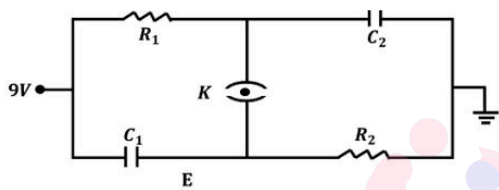
- 1. (A), (B) and (C) only
- 2. (B), (C) and (D) only
- 3. (A), (C) and (D) only
- 4. (A) and (C) only

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.26

In the given circuit, $R_1 = 6\ \Omega$, $R_2 = 3\ \Omega$, $C_1 = 4\ \mu\text{F}$ and $C_2 = 5\ \mu\text{F}$. When key K is closed, the charge stored in C_1 is



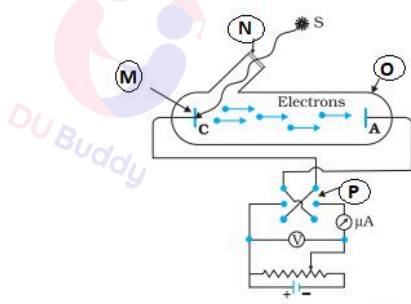
1. $24\ \mu\text{C}$
2. $20\ \mu\text{C}$
3. $12\ \mu\text{C}$
4. $16\ \mu\text{C}$

Options

1. 1
2. 2
3. 3
4. 4

Q.27

In experimental arrangement to study photoelectric effect, here M,N,O and P respectively are



- (A) Commutator
- (B) Evacuated glass tube
- (C) Quartz window
- (D) Photosensitive plate

Choose the **correct** answer from the options given below:

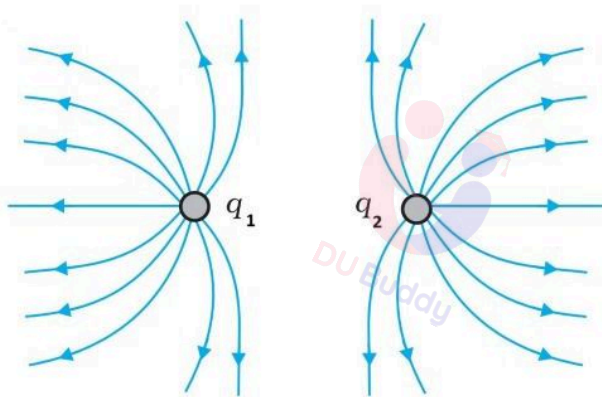
1. (A), (B), (C), (D)
2. (B), (A), (D), (C)
3. (D), (C), (B), (A)
4. (C), (B), (D), (A)

Options

1. 1
2. 2
3. 3
4. 4

Q.28

Identify the correct statement about the electric charges q_1 and q_2 from the figure given below.



1. q_1 is positive and q_2 is negative
2. q_1 is negative and q_2 is positive
3. both q_1 and q_2 are negative
4. both q_1 and q_2 are positive

Options

1. 1
2. 2
3. 3
4. 4

Q.29

In Young's double slit experiment using monochromatic light of wavelength λ the intensity of light at a point on the screen is 20 units, where the path difference is λ . The intensity of light at a point having the path difference $\lambda/4$ is

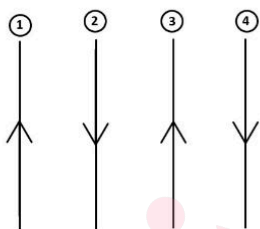
1. 5 units
2. 10 units
3. 20 units
4. 40 units

Options

1. 1
2. 2
3. 3
4. 4

Q.30

Equal currents are flowing in four infinitely long wires. The distance between each other is the same and directions of currents are as shown in the figure. The net force on wire 2 is



Choose the **correct** answer from the options given below:

1. directed towards left
2. equal to that on wire 1 towards right
3. directed towards right
4. zero

Options

1. 1
2. 2
3. 3
4. 4

Q.31

A cell of emf $\mathcal{E} = 10 \text{ V}$ and internal resistance $r = 3 \Omega$ is connected to a uniform wire of length $L = 500 \text{ cm}$ and resistance $R = 3 \Omega$. The drop of potential with length in the wire is:

1. 30 mV/cm
2. 20 mV/cm

3.10 mV/cm

4.5 mV/cm

Options 1. 1

2. 2

3. 3

4. 4

Q.32 Two long straight parallel wires are carrying charges λ_1 and λ_2 per unit length, respectively. The separation between their axes is d . The magnitude of the force exerted on unit length of one wire due to the charge on the other wire is

1. $f = \frac{\lambda_1 \lambda_2}{2\pi\epsilon_0 d}$

2. $f = \frac{\lambda_1 \lambda_2}{4\pi\epsilon_0 d}$

3. $f = \frac{\lambda_1 \lambda_2}{4\pi\epsilon_0 d^2}$

4. $f = \frac{\lambda_1 \lambda_2}{(2\pi\epsilon_0 d)^2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.33 A surface of area $\vec{A} = 2\hat{i} - 3\hat{k}$ exists in a region of electric field $\vec{E} = a\hat{i} + b\hat{j}$. The net flux passing through the surface is

1. $2a - 3b$

2. $3a - 2b$

3. $2a$

4. $-3b$

Options 1. 1

2. 2

3. 3

4. 4

Q.34 An electric bulb rated as 220 W, 110 V is used in a circuit having a 220 V supply.

For the bulb to deliver 220 W, the resistance to be added in the series is

1. 1.55Ω

2. 110Ω

3. 165Ω

4. 220Ω

Options 1. 1

2. 2

3. 3

4. 4

Q.35 Electromagnetic wave carries

1. Charge and momentum

2. Charge and wavelength

3. Energy and momentum

4. Charge and energy

Options 1. 1

2. 2

3. 3

4. 4

Q.36 If the value of the refractive index of a denser medium with respect to the rarer medium is 1.414, then the critical angle is:

1. 30°

2. 45°

3. 60°

4. 75°

Options 1. 1

2. 2

3. 3

Q.37 Two charges of equal magnitude and at distance r exert a force F on each other. If each charge is doubled and the distance between them is halved, then the new force acting on each charge will be

1. $4F$
2. $16F$
3. $2F$
4. $12F$

Options 1. 1

2. 2
3. 3
4. 4

Q.38 In a nuclear fission reaction given below, the three neutrons are released. The value of x and y are



1. $x = 147, y = 36$
2. $x = 144, y = 36$
3. $x = 147, y = 33$
4. $x = 144, y = 33$

Options 1. 1

2. 2
3. 3
4. 4

Q.39 A Fraunhofer diffraction pattern due to a single slit of width 0.2 mm is being obtained on a screen placed at a distance of 2 m from the slit. The first minimum lies at 5 mm on either side of the central maximum on the screen. The wavelength of light is:

1. $4700 \text{ \AA}$
2. $5000 \text{ \AA}$
3. $6000 \text{ \AA}$
4. $6800 \text{ \AA}$

Options 1. 1

2. 2
3. 3
4. 4

Q.40 A 12 pF capacitor is connected to a 50 V battery. The electrostatic energy stored in the capacitor is

1. $2.5 \times 10^{-6} \text{ J}$
2. $3.5 \times 10^{-8} \text{ J}$
3. $2.5 \times 10^{-8} \text{ J}$
4. $1.5 \times 10^{-8} \text{ J}$

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Identify the correct statements.

- (A) A concave mirror is preferred over a plane mirror for the purpose of shaving.
- (B) The absolute refractive index of a medium can be less than unity.
- (C) Snell's law gives the relative refractive index of a medium.
- (D) When a convex lens is placed inside a transparent medium of refractive index greater than that of its own material, it behaves as a diverging lens.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (B), (C) and (D) only
3. (A), (C) and (D) only
4. (A) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.42 Arrange the following electromagnetic waves in their increasing order of energy.

- (A) Gamma rays
- (B) X-rays
- (C) Microwaves

(D) Infrared waves

(E) Ultraviolet rays

Choose the correct answer from the options given below:

1. (A), (B), (C), (D), (E)

2. (E), (D), (C), (B), (A)

3. (C), (D), (E), (B), (A)

4. (C), (D), (B), (A), (E)

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Due to relative motion of a magnet with respect to coil, an emf is induced in the coil in accordance with

1. Ampere's circuital law

2. Faraday's law

3. Gauss's law

4. Biot-savart law

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Identify the correct statements among the following.

(A) At 0 K, the resistance of a semiconductor is infinite.

(B) In an n-type semiconductor, the electron concentration is approximately equal to the concentration of donor atoms.

(C) In p-type semiconductors, the acceptor levels lie near the conduction band.

(D) In n-type semiconductors, the charge carriers are both electrons and holes.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only

2. (B), (C) and (D) only

3. (A), (B) and (D) only

4. (A) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.45 The output frequency of a full-wave rectifier, If the input frequency is 60 Hz is

1. 30 Hz

2. 60 Hz

3. 90 Hz

4. 120 Hz

Options 1. 1

2. 2

3. 3

4. 4

Q.46 A small square loop of wire of side l is placed inside a large square loop of wire of side L ($L \gg l$).

The loops are co-planar and their centres coincide. The mutual inductance of the system is proportional to

1. $\frac{l}{L}$

2. $\frac{l^2}{L}$

3. $\frac{L}{l}$

4. $\frac{L^2}{l^2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.47 If $\vec{M}$ is the magnetic moment and $\vec{B}$ is the magnetic field, then the torque is given by

1. $\vec{M} \cdot \vec{B}$

2. $\frac{|\vec{M}|}{|\vec{B}|}$

3. $\vec{M} \times \vec{B}$

4. $\frac{|\vec{B}|}{|\vec{M}|}$

Options

1. 1
2. 2
3. 3
4. 4

Q.48 Two resistance wires are of same material having length, l and $2l$ and area of cross-section, $4A$ and A respectively. When they are connected in parallel, their equivalent resistance can be expressed as

1. $\frac{2}{9} \left(\rho \frac{l}{A} \right)$
2. $\frac{4}{9} \left(\rho \frac{l}{A} \right)$
3. $\frac{9}{2} \left(\rho \frac{l}{A} \right)$
4. $\frac{9}{4} \left(\rho \frac{l}{A} \right)$

Options

1. 1
2. 2
3. 3
4. 4

Q.49 The angular momentum of an electron in the third orbit of a hydrogen atom is
(Given: $h = 6.6 \times 10^{-34}$ Js)

1. 1.15×10^{-34} Js
2. 2.15×10^{-34} Js
3. 3.15×10^{-34} Js
4. 4.12×10^{-19} Js

Options

1. 1
2. 2
3. 3
4. 4

Q.50 The distance between an object and its virtual image of magnification $\frac{1}{2}$ as produced by a lens is 20 cm. The focal length of the lens is

1. + 40 cm
2. - 20 cm
3. + 20 cm
4. - 40 cm

Options

1. 1
2. 2
3. 3
4. 4